

Sri Satya Venkata Vinay Mahesh Patnala

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PROFESSIONAL SUMMARY

Robotics and Electronics Engineer with an MSc in Robotics from Heriot-Watt University's Edinburgh Centre for Robotics and a B. Tech in Electronics and Communication Engineering. Independently designed and built a novel automated inspection system from scratch as sole researcher handling every phase from concept and data collection through deep learning model training and defect quantification. Hands-on experience across computer vision, 3D reconstruction, embedded firmware, autonomous robotic systems, and human-robot interaction. Driven by genuine curiosity, a commitment to completing every piece of work to the highest standard, and the ability to learn any new tool or domain quickly. Takes full ownership, manages competing priorities without dropping quality, and works effectively alone or in a team. Published conference author. Affiliated researcher, Edinburgh Centre for Robotics.

EDUCATION

MSc Robotics

Jan 2025 – May 2026

Heriot-Watt University, Edinburgh — Edinburgh Centre for Robotics (ECR)

- Dissertation: Defect Detection in Complex Environments using NeRF and 3D Gaussian Splatting — supervised by Dr. Maria Koskinopoulou and Dr. Abderrahim Halimi.
- Modules: Robotic Mechanical Systems, Machine Vision, Robotic System Science, Intelligent Robotics, Human-Robot Interaction, Embedded Software.
- Active member in Heriot-Watt Maker Society prototyped physical components using 3D printers, laser cutters, and resin printers.

B. Tech Electronics and Communication Engineering

Jun 2018 – Jul 2022

Vel Tech Rangarajan Dr. Sagunthala R&D Institute, Chennai, India

- Published: "Design of Printed Antenna for Next Generation Wi-Fi Devices" International Conference on Emerging Technologies, Chennai, May 2022.
- Core focus: embedded systems, Arduino, signal processing, AI fundamentals, and hardware prototyping.

SKILLS

- **Computer Vision & 3D:** NeRF, 3D Gaussian Splatting, COLMAP, Nerfstudio, OpenCV, Point Cloud Processing, Chamfer Distance, ICP, ffmpeg.
- **Deep Learning & AI:** PyTorch, CNN Architectures, LLM Integration (Ollama), CUDA, Scikit-learn, Google Colab.
- **Robotics:** ROS, DH Convention, Forward/Inverse Kinematics, Jacobian Methods, HRI Design, Webots.
- **Embedded & Hardware:** ESP32-C3, FreeRTOS, ESP-IDF, Arduino, I2C, SPI, UART, Oscilloscope, 3D Printing, Laser Cutting
- **Programming:** Python, C, MATLAB, JavaScript, Node.js, Bash, Linux, Git, VS Code.
- **Data Science:** Pandas, NumPy, Matplotlib, Supervised ML, DBSCAN, RANSAC, Data Pipelines.
- **Professional:** Independent research, project ownership, deadline management, fast self-directed learning, team collaboration, written and verbal communication.

KEY PROJECTS

Defect Detection via NeRF and 3D Gaussian Splatting — MSc Dissertation

Jan – May 2026

Heriot-Watt University | Supervised by Dr. Maria Koskinopoulou and Dr. Abderrahim Halimi

- Identified a real industrial problem — manual visual inspection of manufactured components is slow and unscalable and built a fully automated replacement pipeline as sole researcher, owning every phase from blank page to completed system.
- Architected the full pipeline: raw video → ffmpeg frame extraction → COLMAP pose estimation → Nerfstudio model training → point cloud export → automated defect quantification, eliminating manual inspection entirely.
- Evaluated two competing AI reconstruction methods head-to-head; selected the superior approach based on evidence and built the final system around it, demonstrating the ability to make informed technical decisions under research conditions.
- Engineered parallel detection strategies: a 2D illumination-normalised heatmap pipeline for visual anomaly localisation, and a 3D weighted Chamfer Distance pipeline for geometric deviation quantification.

4-DOF Autonomous Robotic Arm

2025

Heriot-Watt University | Supervised by Dr. Xianwen Kong

- Designed, 3D-printed, and assembled a complete four-DOF robotic arm, then implemented full forward and inverse kinematics in MATLAB using DH convention and Jacobian analysis.
- Integrated OpenCV for real-time object detection and localisation, enabling autonomous pick-and-place connecting perception, planning, and actuation in one physical system.

CheerMate: HRI Fitness Coach

2025

Heriot-Watt University | Supervised by Dr. Christian Dondrup

- Led software architecture for a Marty V2 fitness coaching robot integrating an Ollama LLM for natural language dialogue with multimodal feedback (speech and physical movement) that adapts dynamically to each user.
- Built QR-code session tracking enabling personalised multi-user management, turning a single demo into a repeatable user-aware product.

ESP32-C3 Embedded Waveform Generator

2025

Heriot-Watt University | Supervised by Dr. Kieran O'Leary

- Developed dual-signal firmware on ESP32-C3 using FreeRTOS and ESP-IDF, running two concurrent signal tasks (DATA and SYNC); verified timing accuracy on an oscilloscope and resolved hardware-software boundary issues discovered during testing.

WORK EXPERIENCE

Python and Data Science Intern

Feb 2024 – Apr 2024

EI Systems Services, India

- Built a supervised ML model to predict insurance claims outcomes end-to-end pipeline from data cleaning through feature engineering, model training, and evaluation on a real business dataset.
- Developed hands-on proficiency in Python, pandas, and matplotlib through applied analysis on commercial datasets.

Customer Service Assistant

Apr 2025 – Present

Pasta Evangelists, Edinburgh, United Kingdom

- Sustained full-time MSc study alongside part-time shift work in a fast-paced regulated environment demonstrating consistent reliability, workload management, and composure under pressure.
- Collaborated within a diverse multilingual team under strict quality standards, building cross-cultural communication skills transferable to any engineering environment.

PUBLICATIONS AND ACHIEVEMENTS

- Conference Paper: "Design of Printed Antenna for Next Generation Wi-Fi Devices" International Conference on Emerging Technologies, Chennai, May 2022.
- Affiliated Researcher, Edinburgh Centre for Robotics (ECR) one of the UK's leading robotics and autonomous systems research centres, in partnership with the University of Edinburgh.

EXTRACURRICULAR ACTIVITIES

- **Contributed** as an active member of Heriot Watt Makers Society used 3D printers, laser cutters and resin printers to prototype physical components for robotics coursework, bridging classroom design and fabrication.
- **Competitor**, EWB UK Design Challenge (Feb 2026): Co-designed a resilient, multi-utility infrastructure network and executed value engineering that reduced pilot costs by \$10.06M (from \$35.1M to \$25.1M) using modular and optimized assets.
- **Navigated** a full cross-country academic and professional transition from India to UK adapting to new research environment, institution, and engineering culture while delivering MSC level research output.